

MOBILE PHONE PRICE CATEGORY PREDICTION

Using Random Forest Classification

Project Type: Machine Learning / Classification

Primary Algorithm: Random Forest Classifier

Project Start Date: 17-Sep

Business Requirement

A mobile-phone company wants to develop a machine-learning solution to classify mobile phones into different price categories based on their technical specifications. The system should analyze specifications such as RAM, internal storage, battery capacity, camera quality, screen resolution, processor-related features, connectivity and other hardware characteristics to predict the appropriate price category of a mobile phone.

Your responsibility as Team Lead is to coordinate the team, distribute the tasks, monitor progress, identify blockers, review completed work and ensure that all assigned deliverables are completed within the given timeline.

Detailed Work Breakdown

Date	Task ID	Task	Detailed Description	Priority	Expected Deliverable
17-Sep	TA-001	Requirement Understanding	Understand the business problem, objective and expected outcome of mobile phone price classification	High	Requirement Notes
17-Sep	TA-002	Dataset Understanding	Study dataset structure, number of records, columns and data types	High	Dataset Understanding Report
17-Sep	TA-003	Feature Identification	Identify mobile-phone specifications relevant for price classification	High	Feature List
17-Sep	TA-004	Target Identification	Identify the dependent/target variable representing the mobile phone price category	High	Target Variable
17-Sep	TA-005	Data Quality Check	Check missing values, duplicate records, incorrect data types and unusual values	High	Data Quality Report
17-Sep	TA-006	Team Task Allocation	Divide activities among team members and assign ownership	High	Team Task Allocation
18-Sep	TA-007	Data Cleaning	Handle missing values, duplicate records and data inconsistencies	High	Cleaned Dataset
18-Sep	TA-008	Exploratory Data Analysis	Analyze relationships between mobile specifications and price categories	High	EDA Notebook
18-Sep	TA-009	Data Visualization	Create meaningful graphs/charts to identify patterns among price categories	Medium	Visualization Results
18-Sep	TA-010	Feature Preparation	Prepare numerical and categorical features for machine-learning model development	High	ML-ready Dataset
19-Sep	TA-011	Train-Test Split	Divide the dataset into training and testing datasets	High	Train/Test Data
19-Sep	TA-012	Random Forest Classification	Develop and train a Random Forest Classification model	High	Trained Random Forest Model
19-Sep	TA-013	Price Category Prediction	Predict the price category of mobile phones using the trained model	High	Prediction Results
19-Sep	TA-014	Model Evaluation	Evaluate model performance using suitable classification metrics	High	Evaluation Report
19-Sep	TA-015	Feature Importance	Identify which mobile specifications contribute most to price-category predictions	Medium	Feature Importance Results

Date	Task ID	Task	Detailed Description	Priority	Expected Deliverable
19-Sep	TA-016	Documentation & Team Review	Document methodology, observations, results and review each member's contribution	High	Project Documentation & Review Checklist
19-Sep	TA-017	Final Demo	Present the project, dataset, Random Forest approach, results and individual contributions	High	Team Presentation

Project Objective

The main objective is to develop a machine-learning model that can classify a mobile phone into an appropriate price category based on its specifications.

Example input: RAM = 8 GB, Storage = 128 GB, Battery = 5000 mAh, Screen Size = 6.5 inch, Rear Camera = 50 MP, Front Camera = 16 MP, 5G = Yes.

Example output: Predicted Price Category → High. The actual prediction depends on the dataset and trained model.

Dataset Requirements

Column	Description	Type
RAM	RAM capacity	Numerical
Storage	Internal storage	Numerical
Battery	Battery capacity	Numerical
Screen_Size	Display size	Numerical
Rear_Camera	Rear camera megapixels	Numerical
Front_Camera	Front camera megapixels	Numerical
Processor_Speed	Processor speed	Numerical
Cores	Number of processor cores	Numerical
Weight	Phone weight	Numerical
4G	4G availability	Categorical
5G	5G availability	Categorical
Bluetooth	Bluetooth availability	Categorical
WiFi	Wi-Fi availability	Categorical
Price_Category	Price category	Target

Target Variable

Price_Category — example classes: Low, Medium, High, Premium. The exact categories should be based on the selected dataset.

Independent Variables

RAM, Storage, Battery, Screen_Size, Rear_Camera, Front_Camera, Processor_Speed, Cores, Weight, 4G, 5G, Bluetooth and WiFi.

Machine Learning Algorithm

Random Forest Classification

The Random Forest algorithm creates multiple decision trees and combines their outputs to determine the predicted class.

```
from sklearn.ensemble import RandomForestClassifier model = RandomForestClassifier( n_estimators=100, random_state=42 ) model.fit(X_train, y_train)
```

Project Workflow

MOBILE PHONE DATASET → Requirement Analysis → Dataset Understanding → Data Quality Check → Data Cleaning → Exploratory Data Analysis → Data Visualization → Feature Preparation → Train-Test Split → Random Forest Classifier → Model Training → Price Classification → Model Evaluation → Feature Importance → Final Results → Final Demo

EDA / Visualization Requirements

1. Price Category Distribution — bar chart showing the number of phones in each category.
2. RAM vs Price Category — analyze whether higher RAM is associated with different price categories.

- 3. Storage vs Price Category — analyze storage patterns across categories.
- 4. Battery vs Price Category — compare battery capacities across categories.
- 5. Camera vs Price Category — analyze camera specifications across categories.
- 6. Correlation Heatmap — identify relationships between numerical specifications.
- 7. Feature Importance — display the features that contribute most to Random Forest classification.

Model Evaluation

Because this is a classification problem, use classification metrics rather than regression metrics.

Metric	Purpose
Accuracy	Percentage of correctly classified observations.
Precision	Measures how many predicted instances of a class are actually that class.
Recall	Measures how many actual instances of a class were correctly identified.
F1-Score	Harmonic mean of precision and recall.
Confusion Matrix	Shows correct and incorrect predictions for each class.

```
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score, confusion_matrix
accuracy = accuracy_score(y_test, y_pred)
precision = precision_score(y_test, y_pred, average='weighted')
recall = recall_score(y_test, y_pred, average='weighted')
f1 = f1_score(y_test, y_pred, average='weighted')
cm = confusion_matrix(y_test, y_pred)
```

Team Responsibilities

Member	Responsibility
Member 1	Requirement & Dataset Understanding
Member 2	Data Cleaning & Data Quality
Member 3	EDA & Visualization
Member 4	Feature Preparation & Train-Test Split
Member 5	Random Forest, Evaluation & Feature Importance
Team Lead	Coordination, Integration, Review & Final Demo

Team Task Tracking

Task ID	Assigned To	Status	Remarks
TA-001	Member / Team Lead	Pending	—
TA-002	Member / Team Lead	Pending	—
TA-003	Member / Team Lead	Pending	—
TA-004	Member / Team Lead	Pending	—
TA-005	Member / Team Lead	Pending	—
TA-006	Member / Team Lead	Pending	—
TA-007	Member / Team Lead	Pending	—
TA-008	Member / Team Lead	Pending	—
TA-009	Member / Team Lead	Pending	—
TA-010	Member / Team Lead	Pending	—
TA-011	Member / Team Lead	Pending	—
TA-012	Member / Team Lead	Pending	—
TA-013	Member / Team Lead	Pending	—

Task ID	Assigned To	Status	Remarks
TA-014	Member / Team Lead	Pending	—
TA-015	Member / Team Lead	Pending	—
TA-016	Member / Team Lead	Pending	—
TA-017	Member / Team Lead	Pending	—

Expected Final Project Files

```
Mobile-Phone-Price-Prediction/ ■ ■■■ dataset.csv ■■■ Mobile_Phone_Price_Prediction.ipynb ■■■
cleaned_dataset.csv ■■■ model.pkl ■■■ EDA_Report.pdf ■■■ Project_Report.pdf ■■■ requirements.txt ■■■
README.md
```

Main Python Libraries

pandas, numpy, matplotlib, seaborn, scikit-learn, joblib

Final Prediction Example

Input: RAM 8 GB; Storage 128 GB; Battery 5000 mAh; Screen Size 6.5 inch; Rear Camera 50 MP; Front Camera 16 MP; Processor Speed 2.4 GHz; Cores 8; 5G Yes.

Output: Predicted Price Category: High. This is an example only; actual results depend on the trained model.

Team EOD Reporting

Each team member should submit an individual EOD report. The Team Lead should additionally submit the following Team EOD Summary:

Item	Details
Team	Team A
Date	_____
Total Members	_____
Tasks Assigned	_____
Tasks Completed	_____
Tasks In Progress	_____
Tasks Pending	_____
Blockers	_____
Files Submitted	_____
Plan for Next Day	_____

Example — 17-Sep

Item	Details
Team	Team A
Date	17-Sep
Total Members	5
Tasks Assigned	6
Tasks Completed	5
Tasks In Progress	1
Tasks Pending	0
Blockers	None
Files Submitted	Requirement Notes, Dataset Report, Feature List
Plan for Next Day	Complete data cleaning, EDA and visualization

Project Classification Note

This project is a **classification** problem, not a regression problem. The main model is **RandomForestClassifier**. The model predicts a category such as Low, Medium, High or Premium rather than an exact rupee price.